



Aqsa Khalid
Lecturer
School of Computing

CS3002 Information Security



Advanced Encryption Standard

Advanced Encryption Standard

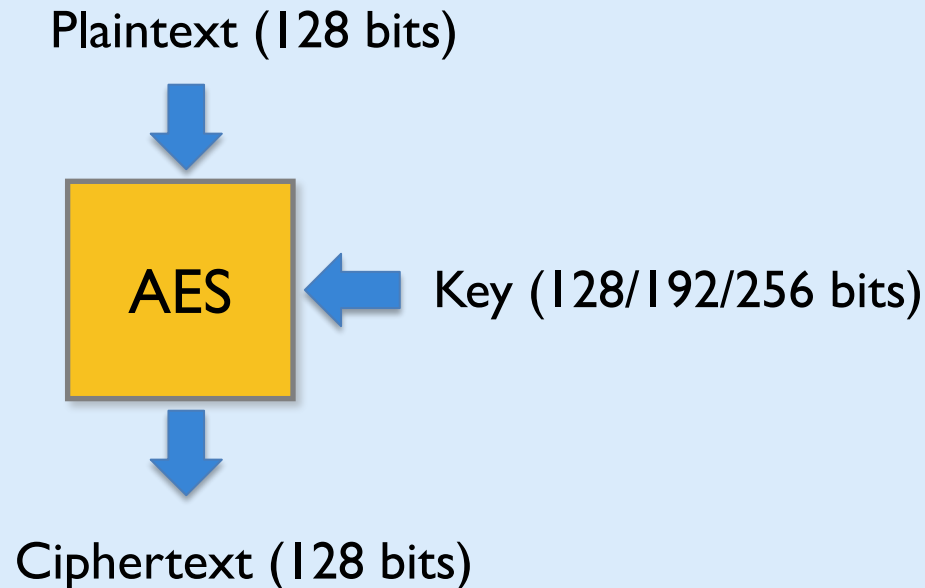


- In 1997, NIST called for proposals for a new cipher to be standardized
 - goals: efficiency, security, HW/SW suitability
 - key length 128, 192, 256 bits
- They selected Rijndael cipher in Nov 2001 and declared it as 'Advanced Encryption Standard'.
- It is also a block cipher. Uses 128 bit data block & 128/192/256 bit keys
- Now its implementations are widely available

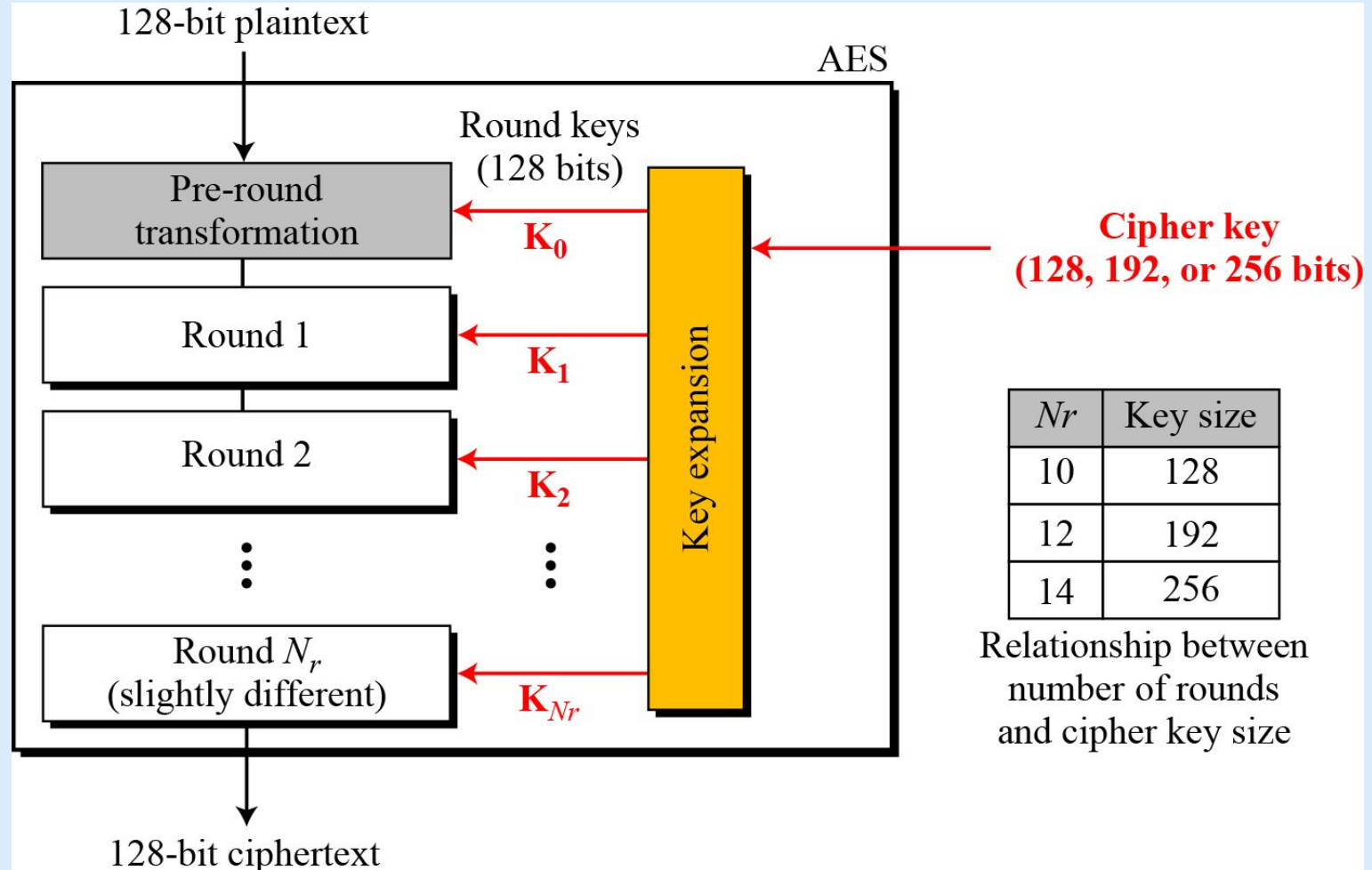
The AES Cipher - Rijndael



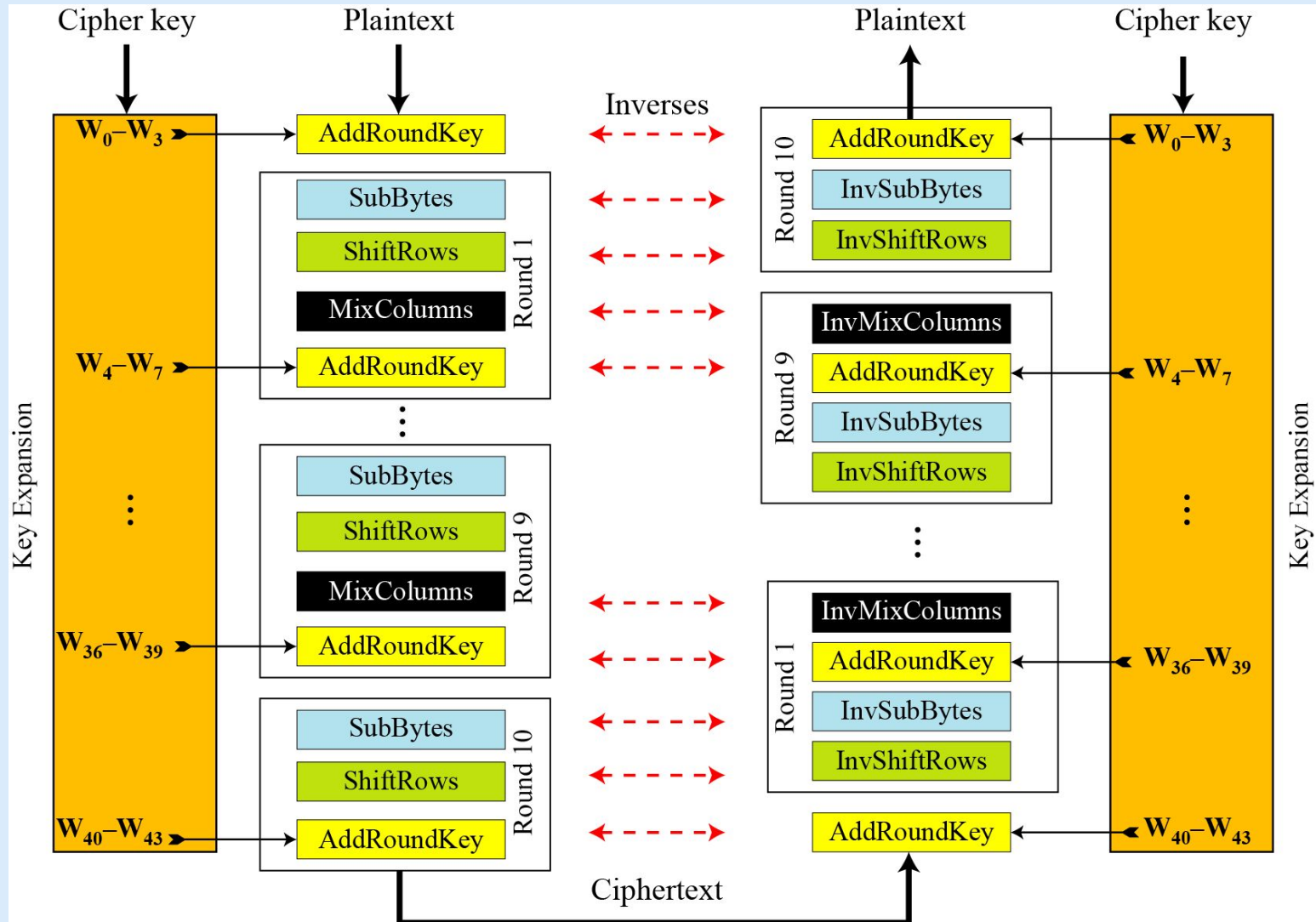
- An iterative cipher with several rounds
 - processes data as block of 4 columns of 4 bytes (128 bits)
 - operates on entire data block in every round



Multiple rounds



Overall Structure

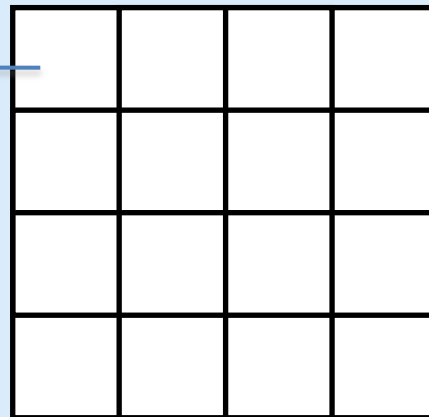


Structure of 128-bit block

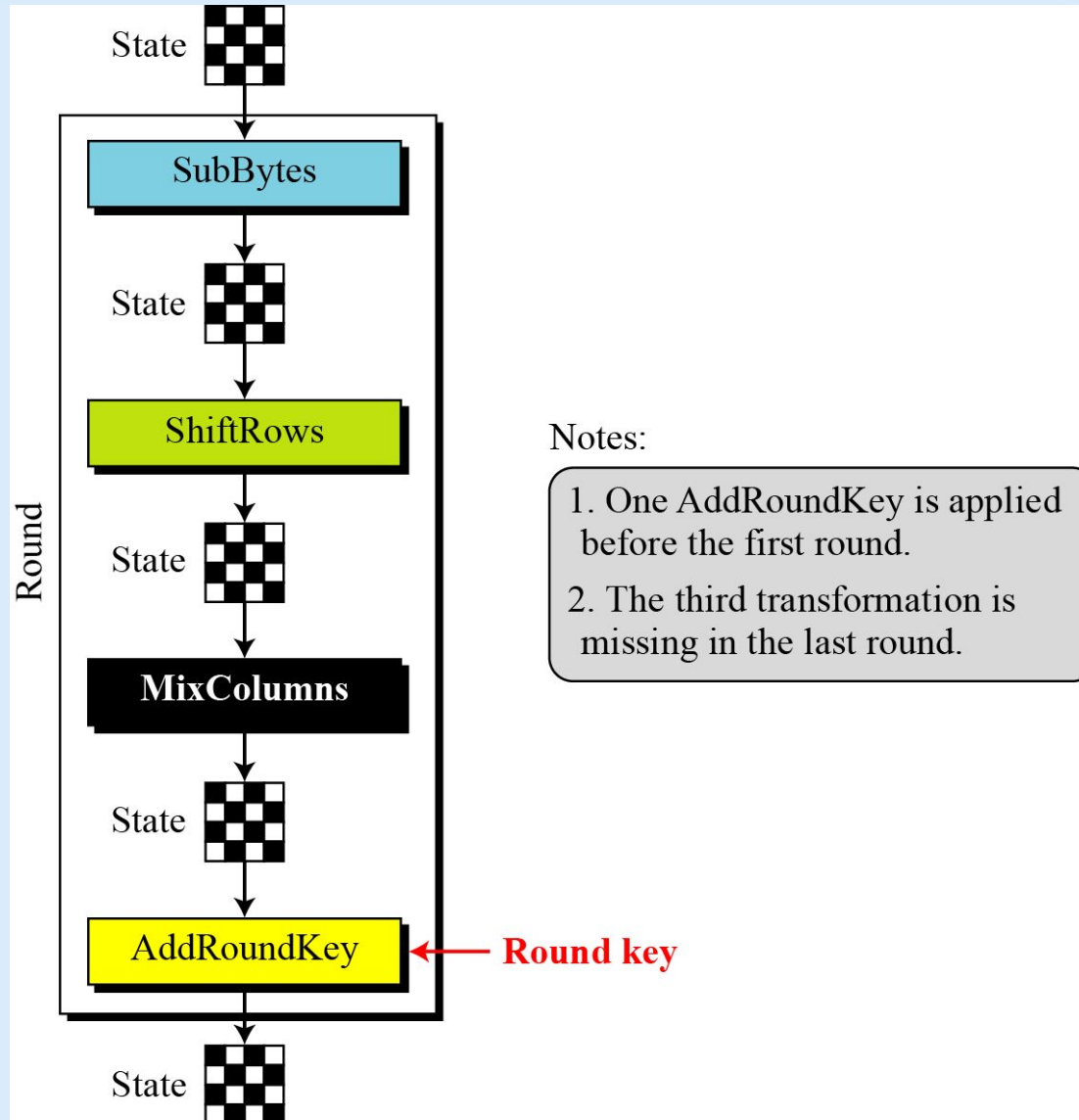


- Data block viewed as table of bytes
- Represented as 4 by 4 matrix of bytes.

1 byte



Details of each round

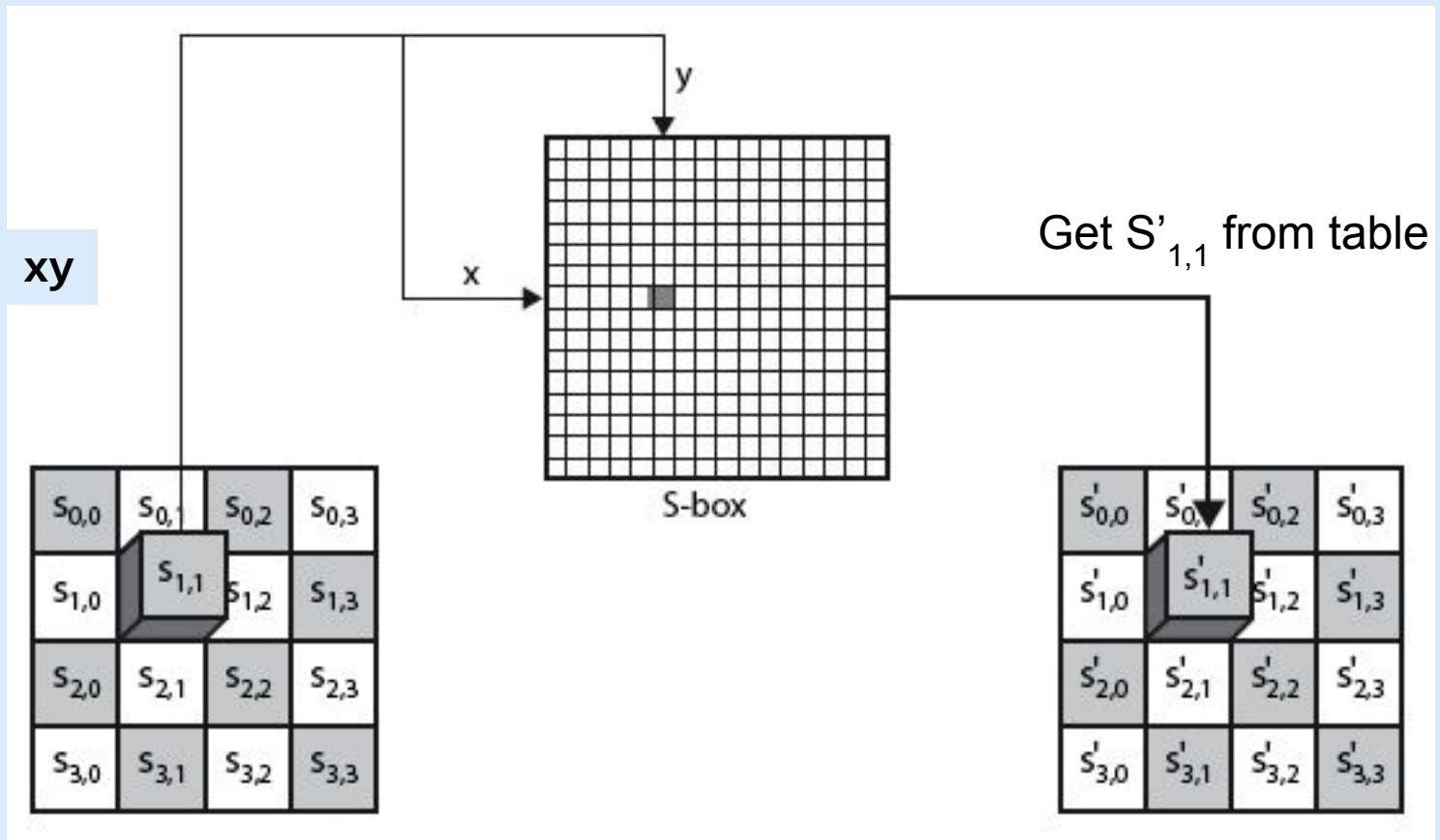


SubBytes Operation



The SubBytes operation involves 16 independent byte-to-byte substitutions.

* Let $S_{1,1} = xy$



* Interpret a byte as two hexadecimal digits xy

SubBytes Table



Implemented by table lookup

		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
x	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
	D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

InvSubBytes Table

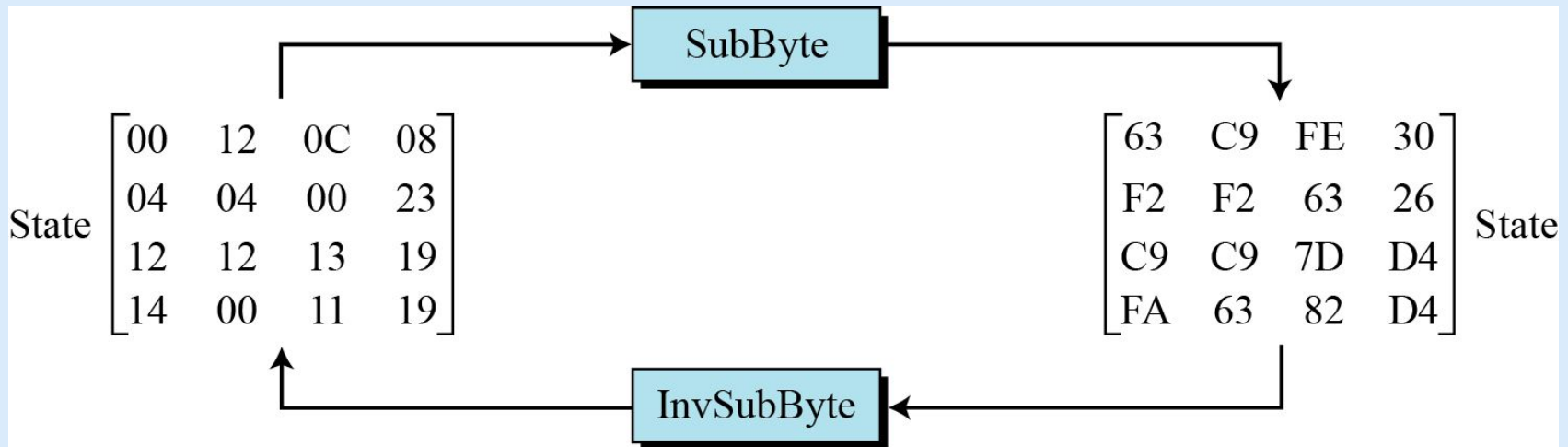


		y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
x	0	52	09	6A	D5	30	36	A5	38	BF	40	A3	9E	81	F3	D7	FB
	1	7C	E3	39	82	9B	2F	FF	87	34	8E	43	44	C4	DE	E9	CB
	2	54	7B	94	32	A6	C2	23	3D	EE	4C	95	0B	42	FA	C3	4E
	3	08	2E	A1	66	28	D9	24	B2	76	5B	A2	49	6D	8B	D1	25
	4	72	F8	F6	64	86	68	98	16	D4	A4	5C	CC	5D	65	B6	92
	5	6C	70	48	50	FD	ED	B9	DA	5E	15	46	57	A7	8D	9D	84
	6	90	D8	AB	00	8C	BC	D3	0A	F7	E4	58	05	B8	B3	45	06
	7	D0	2C	1E	8F	CA	3F	0F	02	C1	AF	BD	03	01	13	8A	6B
	8	3A	91	11	41	4F	67	DC	EA	97	F2	CF	CE	F0	B4	E6	73
	9	96	AC	74	22	E7	AD	35	85	E2	F9	37	E8	1C	75	DF	6E
	A	47	F1	1A	71	1D	29	C5	89	6F	B7	62	0E	AA	18	BE	1B
	B	FC	56	3E	4B	C6	D2	79	20	9A	DB	C0	FE	78	CD	5A	F4
	C	1F	DD	A8	33	88	07	C7	31	B1	12	10	59	27	80	EC	5F
	D	60	51	7F	A9	19	B5	4A	0D	2D	E5	7A	9F	93	C9	9C	EF
	E	A0	E0	3B	4D	AE	2A	F5	B0	C8	EB	BB	3C	83	53	99	61
	F	17	2B	04	7E	BA	77	D6	26	E1	69	14	63	55	21	0C	7D

SubByte Example



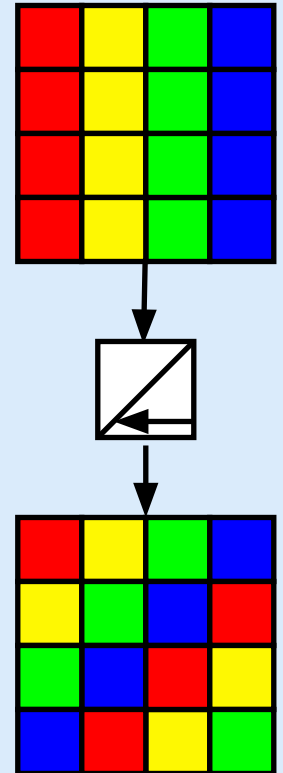
- The SubBytes and InvSubBytes transformations are inverses of each other.



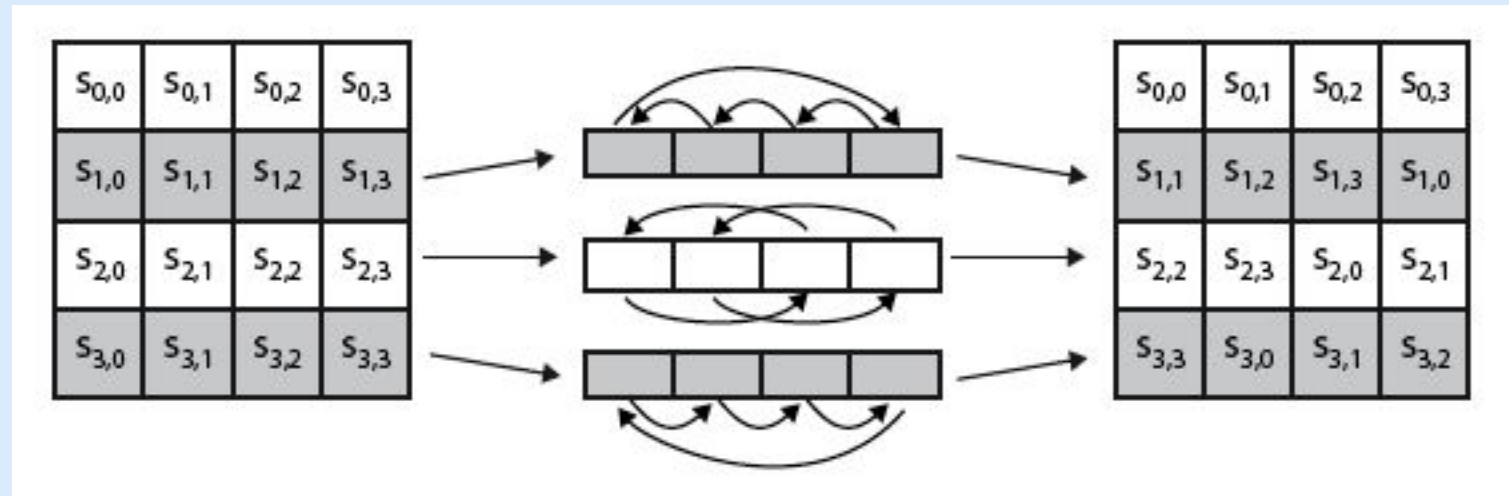
ShiftRows



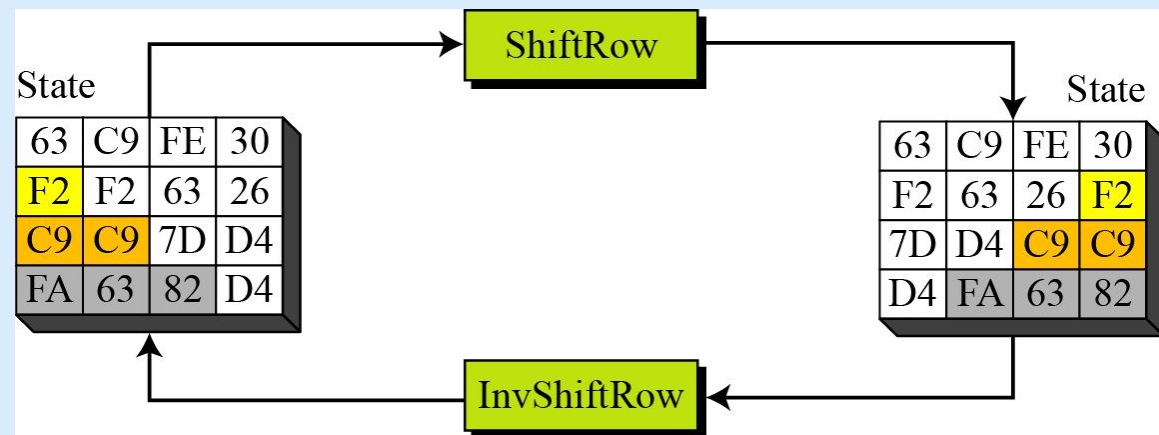
- Shifting permutes the bytes.
- Does a circular byte shift in each row
 - 1st row is unchanged
 - 2nd row does 1 byte circular shift to left
 - 3rd row does 2 byte circular shift to left
 - 4th row does 3 byte circular shift to left
- In encryption, the transformation is called ShiftRows
- In decryption, the transformation is called InvShiftRows and the shifting is to the right



ShiftRows Scheme



Example

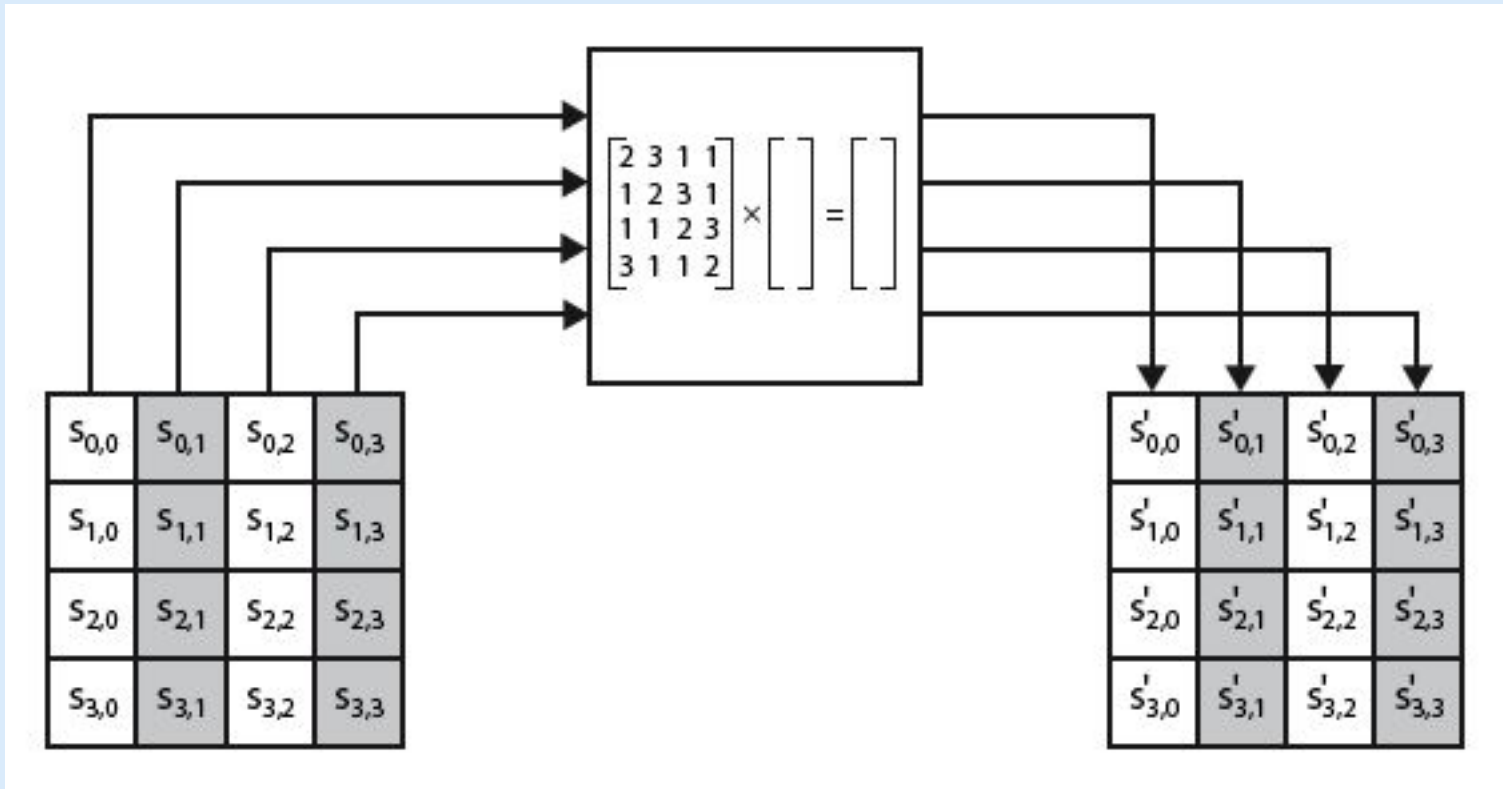


MixColumns



- ShiftRows and MixColumns provide diffusion to the cipher
- In MixColumns, each column is processed separately
- Each byte is replaced by a value dependent on all 4 bytes in the column
- Effectively a matrix multiplication in finite field $GF(2^8)$ using prime polynomial $x^8+x^4+x^3+x+1$
 - Mathematics for this part is beyond course scope

MixColumns Scheme



The MixColumns transformation operates at the column level; it transforms each column of the state to a new column.

MixColumn and InvMixColumn



During decryption, inverse mixing matrix is used

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \xleftrightarrow{\text{Inverse}} \begin{bmatrix} 0E & 0B & 0D & 09 \\ 09 & 0E & 0B & 0D \\ 0D & 09 & 0E & 0B \\ 0B & 0D & 09 & 0E \end{bmatrix}$$

C C^{-1}

AddRoundKey



This transformation XORs state matrix with 128-bits of the round key

- AddRoundKey proceeds one column at a time.
 - adds a round key word to each state column matrix
 - the operation is matrix addition
- Inverse for decryption is identical
 - since XOR is its own inverse, with same keys

AddRoundKey Scheme



$s_{0,0}$	$s_{0,1}$	$s_{0,2}$	$s_{0,3}$
$s_{1,0}$	$s_{1,1}$	$s_{1,2}$	$s_{1,3}$
$s_{2,0}$	$s_{2,1}$	$s_{2,2}$	$s_{2,3}$
$s_{3,0}$	$s_{3,1}$	$s_{3,2}$	$s_{3,3}$

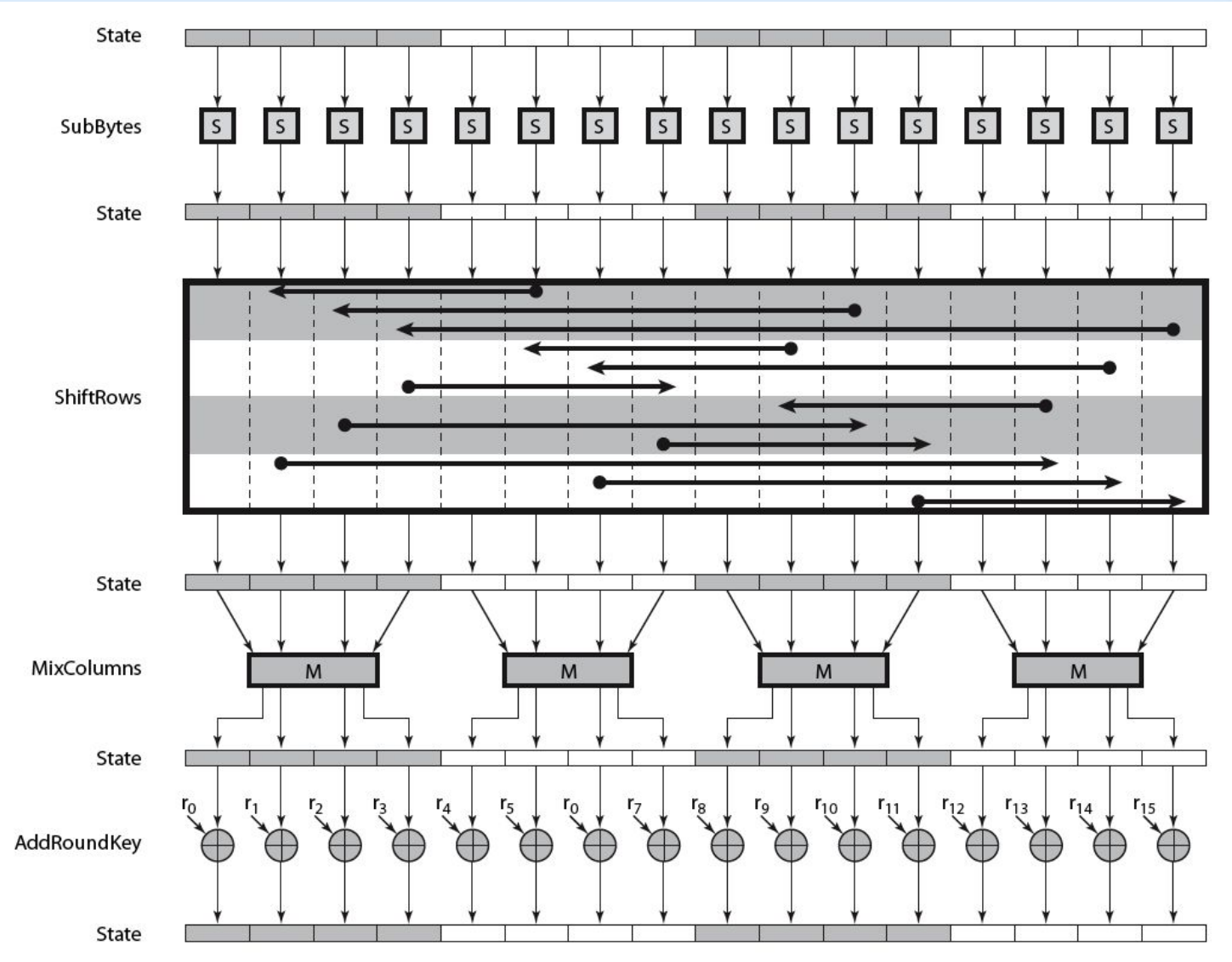
$\oplus$

w_i	w_{i+1}	w_{i+2}	w_{i+3}
-------	-----------	-----------	-----------

$=$

$s'_{0,0}$	$s'_{0,1}$	$s'_{0,2}$	$s'_{0,3}$
$s'_{1,0}$	$s'_{1,1}$	$s'_{1,2}$	$s'_{1,3}$
$s'_{2,0}$	$s'_{2,1}$	$s'_{2,2}$	$s'_{2,3}$
$s'_{3,0}$	$s'_{3,1}$	$s'_{3,2}$	$s'_{3,3}$

AES Round



AES Key Schedule (generating round keys)



- takes 128-bits (16-bytes) key and expands into an array of 44 words (32-bit each)

<i>Round</i>	<i>Words</i>			
Pre-round	$\mathbf{w}_0$	$\mathbf{w}_1$	$\mathbf{w}_2$	$\mathbf{w}_3$
1	$\mathbf{w}_4$	$\mathbf{w}_5$	$\mathbf{w}_6$	$\mathbf{w}_7$
2	$\mathbf{w}_8$	$\mathbf{w}_9$	$\mathbf{w}_{10}$	$\mathbf{w}_{11}$
...	...			
N_r	$\mathbf{w}_{4N_r}$	$\mathbf{w}_{4N_r+1}$	$\mathbf{w}_{4N_r+2}$	$\mathbf{w}_{4N_r+3}$

AES Security



- AES was designed after DES.
 - Most of the known attacks on DES were already tested on AES.
- Brute-Force Attack
 - AES is definitely much more secure than DES due to the larger-size key.
- Statistical Attacks
 - Numerous tests have failed to do statistical analysis of the ciphertext
- Differential and Linear Attacks
 - There are no known differential and linear attacks on AES as yet.

Implementation Aspects



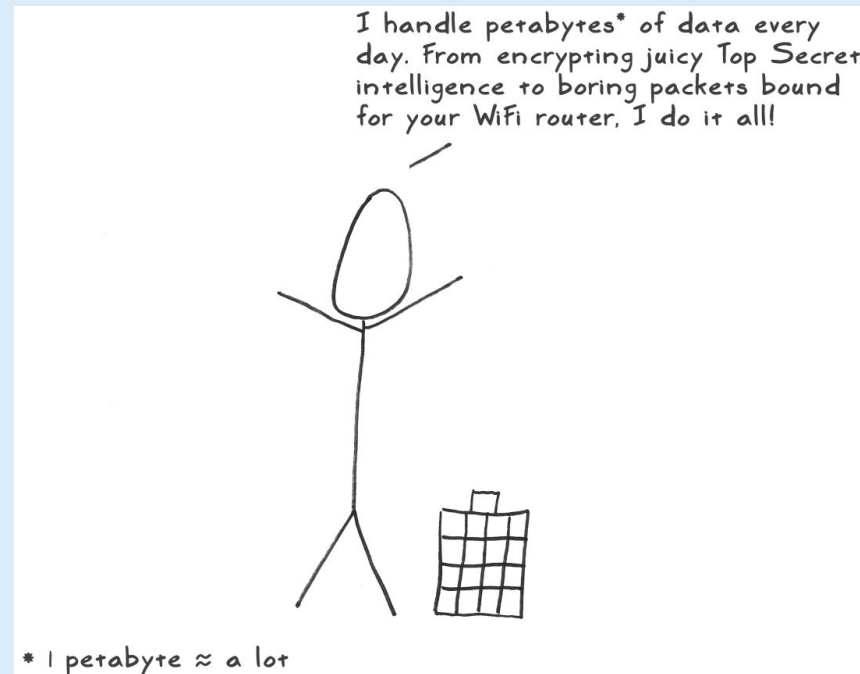
- The algorithms used in AES are so simple that they can be easily and efficiently implemented using cheap processors and a minimum amount of memory.
 - Fast implementation was a key factor in its selection as the AES
- Several modern CPU architectures include AES instructions:
https://en.wikipedia.org/wiki/AES_instruction_set

AES illustrated



A stick figure guide

<http://www.moserware.com/2009/09/stick-figure-e-guide-to-advanced.html>



References



- Stallings CNS, chap 4, 6
- Forouzan, Cryptography & Network Security, chap 5–8
- Confusion, Diffusion and AES steps with reasoning:
<https://youtu.be/C4ATDMLz5wc>
- AES illustrated: a stick figure guide:
<http://www.moserware.com/2009/09/stick-figure-guide-to-advanced.html>
- Block cipher modes of operation – Computerphile:
<https://youtu.be/Rk0NIQfEXBA>